

Full-atlas TCR metadata correction

Sample-aware validated receptor calls were applied to a separate 5.93-million-cell atlas candidate while preserving expression, integration, and the canonical rollback object.

5,933,312 atlas cells	2,155,409 corrected rows
2,270,138 productive AB	1,938,158 paired TRA/TRB

Decision summary

PASS_TCR_H5AD_CANDIDATE: all 17 transaction checks passed. The candidate was renamed from `.partial` only after exact sidecar, unaffected-value, matrix, embedding, and backed-read validation.

Candidate: `/ssd/tnk_phase3/Integrated_dataset/full_atlas/tcr_corrected/integrated_full_atlas.h5ad`

SHA-256: `d32c9d2bdb955b12e1eafbed8322f8cb965cf3a225191e612b53f3d3783480d5`

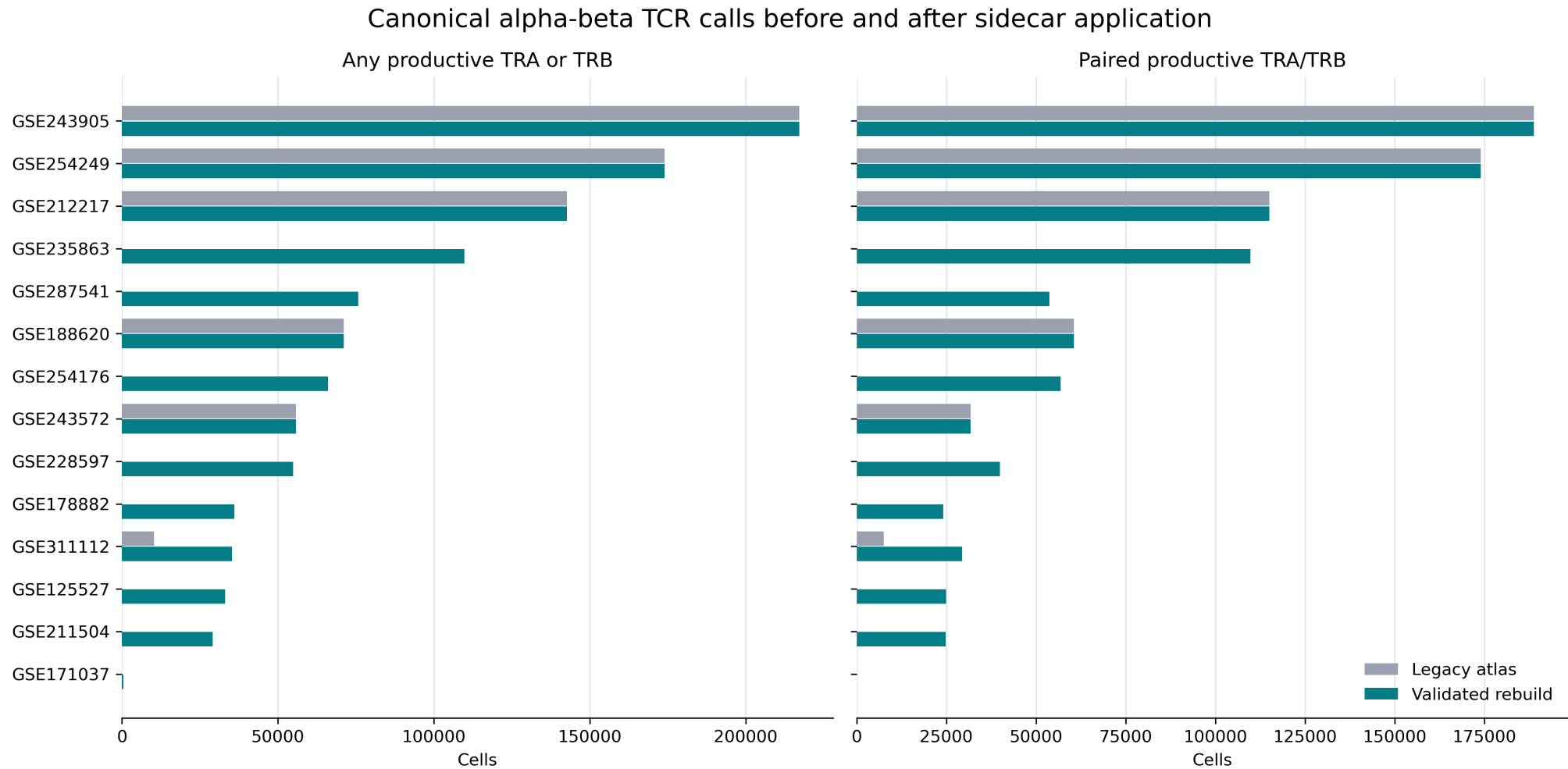
The canonical symlink was not changed. The metadata-corrected input and original canonical atlas remain rollback points.

What changed

For 14 repaired sources, atlas `source_gse_id + original_cell_id` was joined one-to-one to sidecar `source_gse_id + source_obs_name`. Forty-six canonical chain and logical fields were replaced, including empty values that remove stale calls. Twenty-seven provenance fields record coverage, eligibility, join status, UMI/read availability, productive-contig multiplicity, source file, and selected contig.

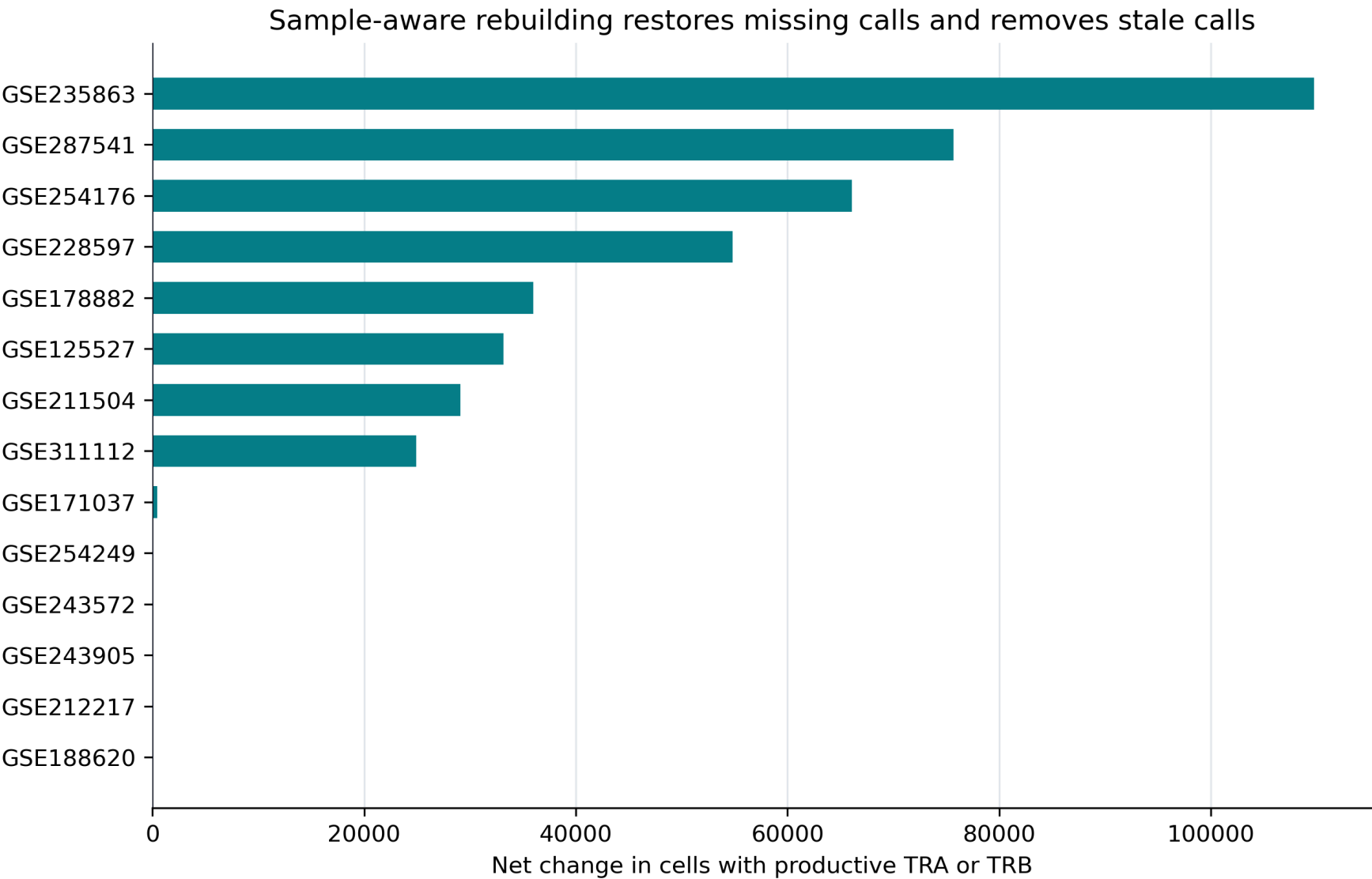
Unavailable UMI/read support remains null, not zero. The 109 atlas-present ambiguous GSE235863 rows are blank and TCR-assignment-ineligible. The 886,462 sidecar rows outside the frozen T/NK atlas were not rescued.

Before and after



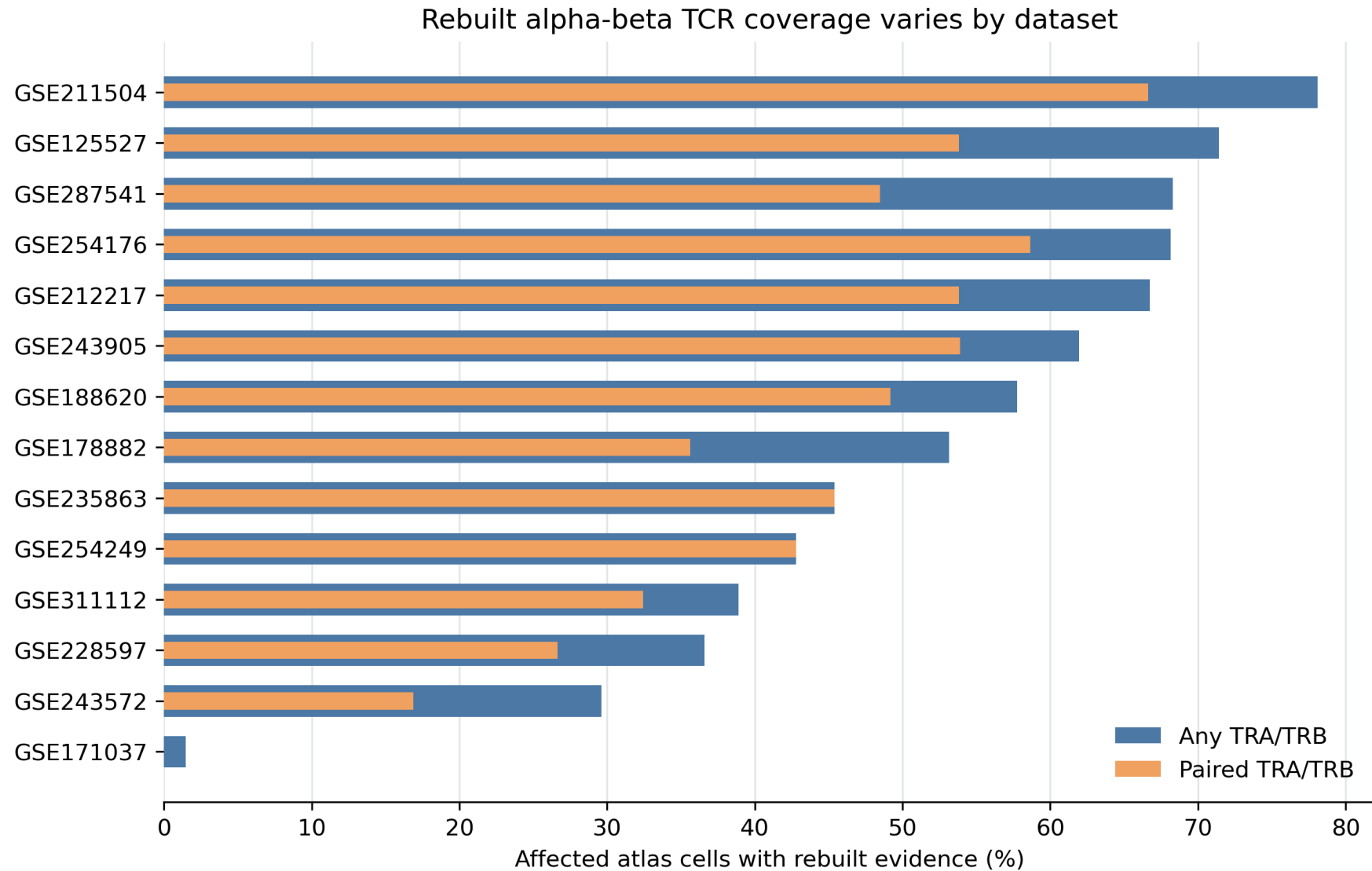
Legacy versus rebuilt productive and paired alpha-beta calls among affected atlas cells.

Where calls changed



Positive values are recovered validated calls; GSE311112 also removes 681 stale productive-alpha-beta assignments.

Dataset coverage



These percentages describe only cells retained in the frozen atlas and should not be interpreted as tissue-level biological abundance.

Source-level audit

Source	Atlas cells	Old any AB	Rebuilt any AB	Old paired	Rebuilt paired	Old-only	Rebuilt-only	Ineligible
GSE125527	46,396	0	33,141	0	24,976	0	33,141	0
GSE171037	31,937	0	472	0	0	0	472	0
GSE178882	67,755	0	36,002	0	24,148	0	36,002	0
GSE188620	123,218	71,160	71,160	60,599	60,599	0	0	0
GSE211504	37,244	0	29,086	0	24,811	0	29,086	0
GSE212217	213,706	142,604	142,604	115,048	115,048	0	0	0
GSE228597	149,812	0	54,819	0	39,925	0	54,819	0
GSE235863	241,766	0	109,741	0	109,741	0	109,741	109
GSE243572	188,217	55,770	55,770	31,767	31,767	0	0	0
GSE243905	350,438	217,164	217,164	188,863	188,863	0	0	0
GSE254176	96,953	0	66,062	0	56,849	0	66,062	0
GSE254249	406,494	174,010	174,010	174,010	174,010	0	0	0
GSE287541	110,874	0	75,707	0	53,761	0	75,707	0
GSE311112	90,599	10,333	35,243	7,590	29,392	681	25,591	0

Affected-source totals: 1,100,981 rebuilt productive-alpha-beta cells and 933,890 rebuilt paired TRA/TRB cells.

Validation matrix

Check	Status
affected tcr values exact	PASS
anndata backed open	PASS
canonical flag consistency affected	PASS
canonical symlink unchanged	PASS
ineligible atlas rows exact	PASS
ineligible rows fail closed	PASS
metadata source state and sha unchanged	PASS
only expected provenance columns added	PASS
protected structure sampled identity	PASS
scvi shape	PASS
sidecar coverage exact	PASS
umap shape	PASS
unaffected tcr values exact	PASS
whole any ab total	PASS
whole paired ab total	PASS
x nnz	PASS
x shape	PASS

Interpretation and next boundary

This transaction repairs receptor metadata; it does not retrain gdTAI, change expression scores, alter cell membership, or rerun scVI, UMAP, or Leiden. Previous NK-boundary and TCR-ground-truth summaries are now stale because they consumed legacy receptor fields.

Next: regenerate TCR-derived truth/control labels and boundary/NK audits from this checksum-bound candidate. Canonical publication remains a separate supervised decision after those reports are reviewed.